

**BIT 4107-Mobile Application**

**Development**

**ENHANCED PROFESSIONAL STUDENT LOGBOOK**

**Student name: KELLY MAROA**

**Admission Number:BIT/2024/49929**

**School/Faculty: COMPUTING AND INFORMATICS**

**Programme:Blood\_Donation\_App**

**Academic Year: THREE**

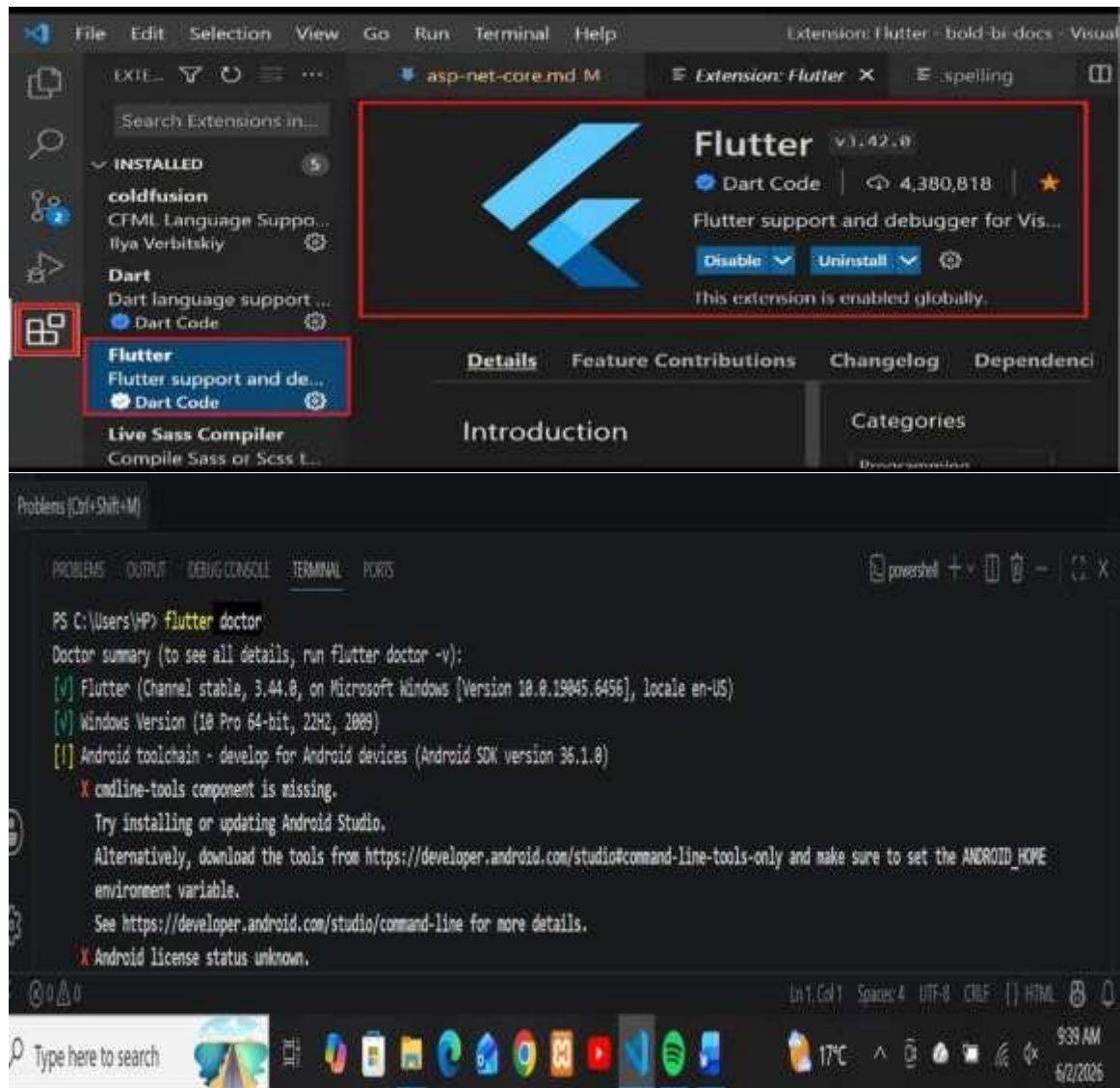
**Semester:TWO**

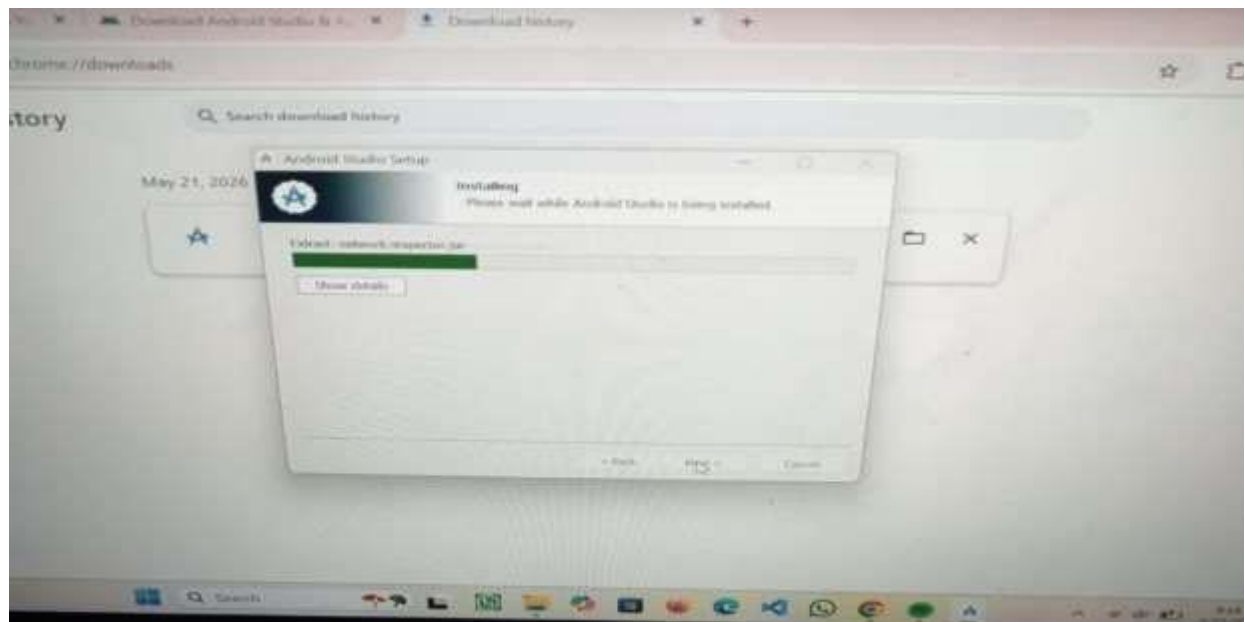
**Lecturer: MR.MICHAEL NYORO**

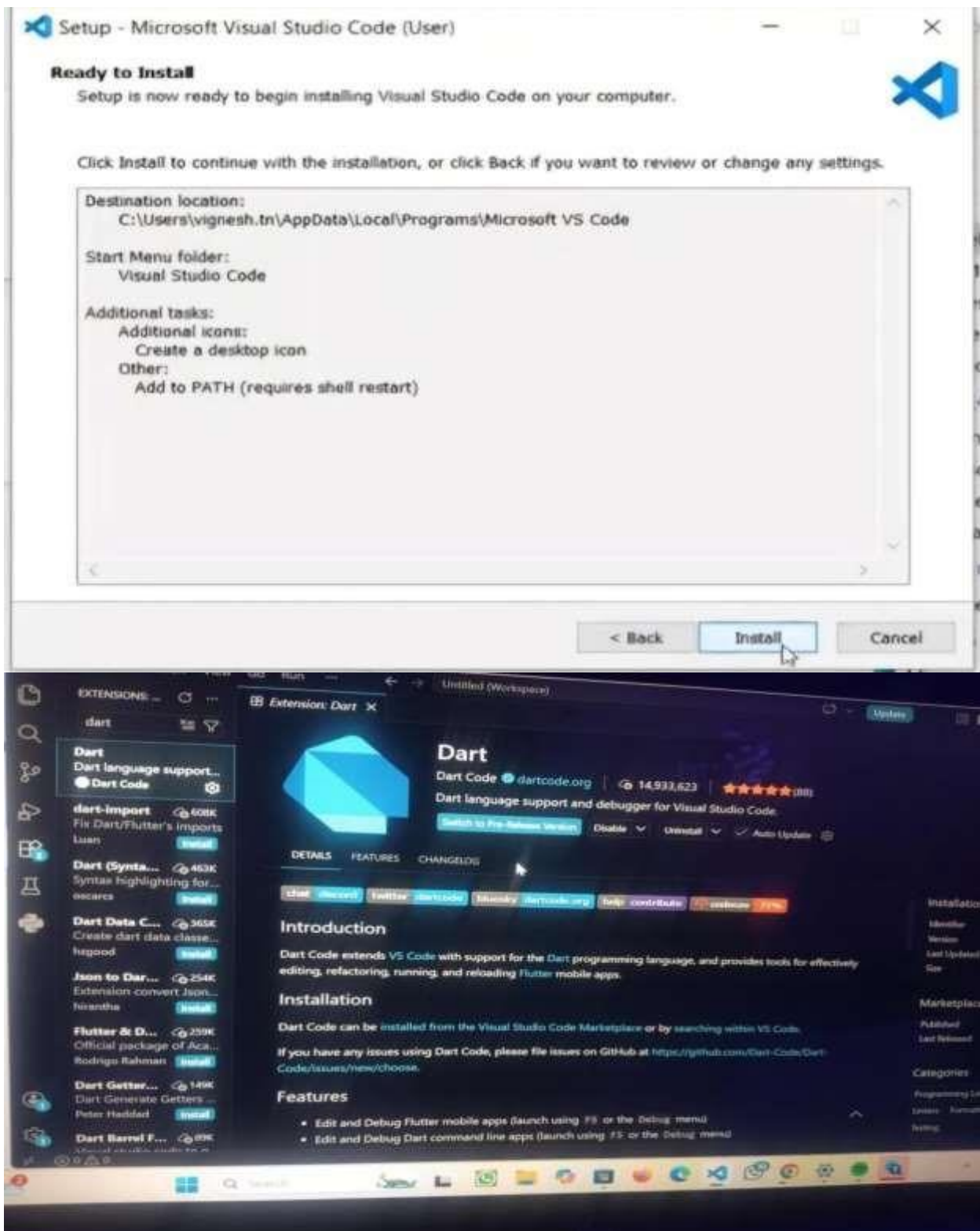
**Phone Number:0725104634**

**WEEK 1: INTRODUCTION TO MOBILE APPLICATION**

**DEVELOPMENT**







TOOLSANDSOFTWAREUSED 1.Android

Studio

2.VS Code

3.Flutter SDK

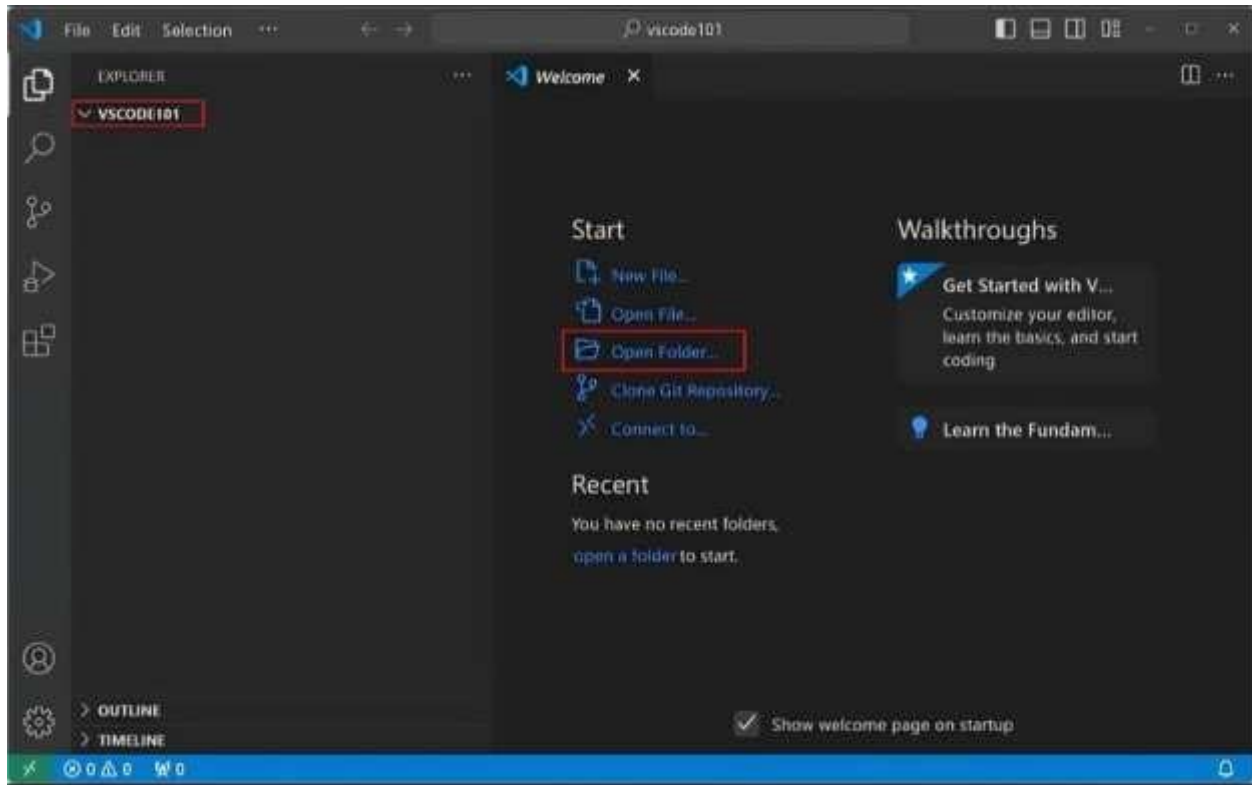
4.Dart

5.Android Phone

## PERSONAL REFLECTION

Installing the tools was challenging at first due to errors and configuration issues, but it improved my problemsolving skills. I gained practical experience in setting up development environments and became more confident in using modern software tools for app development.

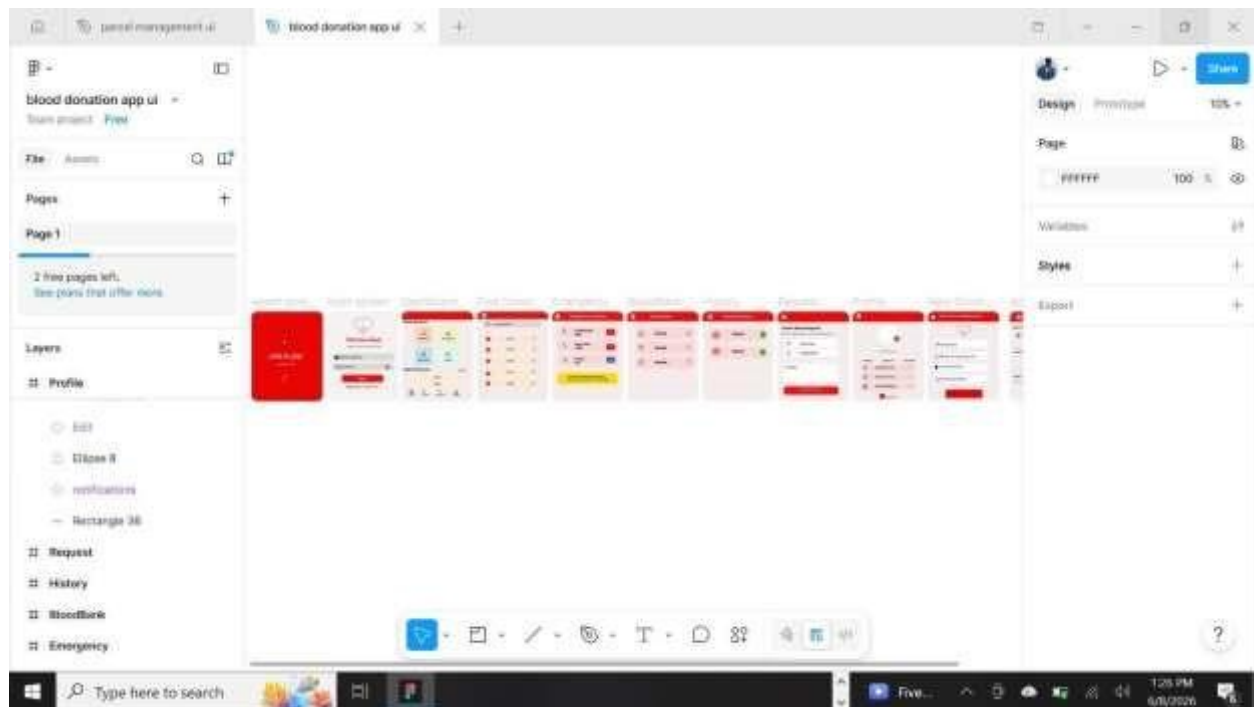
## WEEK 2: MOBILE DEVELOPMENT ENVIRONMENT



## PERSONAL REFLECTION

I set up the mobile development environment by installing and configuring tools like Android Studio, Flutter SDK, and device emulators. This process helped me understand how different components work together, though I faced minor setup challenges that enhanced my troubleshooting and technical skills.

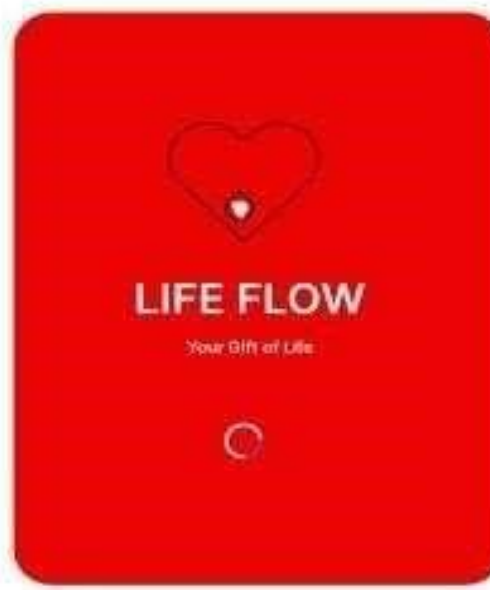
## WEEK 3: USER INTERFACE DESIGN



login screen



spash screen





## Personal reflection

I focused on user interface design for the blood donation app using Figma. I learned how to create and organize screens, design user-friendly layouts, and improve overall system usability. This improved my understanding of UI/UX principles and enhanced my design skills.

Week 4: Event handling and user interaction



login screen



The login screen features a light pink background. At the top center is a heart icon with a blood drop inside. Below it, the text "Welcome Back" is displayed in a bold, dark red font, followed by the subtitle "Login to continue saving lives" in a smaller, grey font. There are two input fields: "Email address" with an envelope icon and "password" with a lock icon and a toggle for visibility. A red "login" button is positioned below the fields. At the bottom, a link "New donor? Register now" is shown in red text.

Heart icon with blood drop

**Welcome Back**

Login to continue saving lives

Email address

password

login

New donor? Register now

Request



The "Request" screen has a red header bar with a back arrow and a notification bell icon. The title "Create Blood Request" is in bold black text, with the instruction "Fill the details below to notify nearby donors" below it. The form includes two dropdown menus for "Blood Group" and "Hospital Name", each with a heart icon. A large text area for "Message" is provided. A red "Broadcast Request" button is at the bottom.

Back arrow

Notification bell

**Create Blood Request**

Fill the details below to notify nearby donors

Blood Group

Hospital Name

Message

Broadcast Request

18:54



## Account Settings

### Appearance



Dark Mode



### Security



### About



1.0.0

### Change Password

Old Password

New Password

Cancel

Save



## Personal reflection

I learned about event handling and user interaction in the blood donation app. I implemented how users interact with the system through buttons, forms, and input fields. This improved my understanding of how applications respond to user actions and enhanced my programming skills.

## Week 5: Data management

19:08



## New Donor Registration



Full Name



Kelly maroa

Blood Group (e.g. O+)



O+

Location/City



Machakos

Phone Number



0785324646

Register as Donor

Hello, Maroakelly2



Search Blood Group



**Anna Itotiah W.**  
Nairobi



**Sharon Kendi.**  
Nakuru



**Faith Mueni.**  
Mombasa



**Roy Gichinga.**  
Kisumu



**Michelle Njogu.**  
Eldoret



**Kelley Maroa**  
Migori



**Nicole Chepkemboi**  
Nandi



Home



Search



Request



Profile



---

## Personal reflection

I focused on data management in the Blood Donation App. I learned how to store, retrieve, and update donor and request information using local storage and databases. This improved my understanding of how data is handled efficiently and enhanced my skills in managing application data.

## Week 6: Networking and APIs

```
1 {
2   "project_info": {
3     "project_number": "596793580149",
4     "project_id": "blood-donation-app-a7042",
5     "storage_bucket": "blood-donation-app-a7042.firebaseioapp"
6   },
7   "client": [
8     {
9       "client_info": {
10        "mobilesdk_app_id": "1:596793580149:android:d5f7eac4ef37ed8144363c",
11        "android_client_info": {
12          "package_name": "com.blood_donation.app"
13        }
14      },
15      "oauth_client": [],
16      "api_key": [
17        {
18          "current_key": "AIzaSyDy62fZd7rrAi_DWKh7gyK9Nb91T9No"
19        }
20      ],
21      "services": {
22        "appinvite_service": {
23          "other_platform_oauth_client": []
24        }
25      }
26    }
27  ],
28   "configuration_version": "1"
29 }
```

A blood donation app requires networking to connect users with donors and hospitals, enable real-time updates, send requests, share location data, and store information on cloud servers for accessibility and communication.

Week 7: Data storage and Database integration in Mobile application

blood donation app - Database

console.firebase.google.com/u/0/project/blood-donation-app-47042/firestore/databases/default/data/~3Fdonors~27Marakelly1@gmail.com

firebase

Search for products

Project Overview

Settings

Project shortcuts

- Firestore
- Realtime Database
- Authentication
- Storage
- Phone Verification

Product categories

Spark  
No cost (\$0/month)

Upgrade

blood donation app - Cloud Firestore - Database (default)

donors

Marakelly1@gmail.com

+ Start collection

+ Add document

+ Start collection

+ Add field

donors

requests

Marakelly1@gmail.com

Huenifaith@gmail.com

huenifaith@gmail.com

bloodGroup: "O-"

email: "Marakelly1@gmail.com"

isActive: true

location: "Unknown"

name: "Marakelly1"

phone: ""

profilePic: "data/user/0/com.blood.donation.app/pc2e36-4b03-b98b-41e446329b82/1000x77255.jpg"

Phone number verification is a fast and secure method for verifying user phone numbers

Type here to search

19°C

11:36 AM

6/17/2024



blood donation app - Database

console.firebase.google.com/u/0/project/blood-donation-app-a7042/firestore/databases/-default-/data/-2frequests-2f56MW0kouZrbP2Wcwn8...

blood donation app • Cloud Firestore • Database (default)

> requests > 56MW0kouZrbP2Wcwn8

| (default)          | requests              | 56MW0kouZrbP2Wcwn8                                                                                                                                                                                                     |
|--------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| + Start collection | + Add document        | + Start collection                                                                                                                                                                                                     |
| docs               | 56UW68kouZrbP2Wcwn8 > | + Add field                                                                                                                                                                                                            |
| requests           | pQciE7JxHQ0RL4dScwkd  | blood: "O+"<br>date: "Just now"<br>location: "Agha Khan Hospital"<br>message: "Emergency"<br>serverTimestamp: "12 June 2026 at 13:03:15 UTC+3"<br>timestamp: "2026-06-12T11:42:03.628035"<br>type: "Emergency Request" |

Type here to search

19°C

11:37 AM 6/12/2026

blood donation app - Auth

console.firebase.google.com/u/0/project/blood-donation-app-a7042/authentication/users

blood donation app •

## Authentication

Users Sign-in method Templates Usage Settings Extensions

The following authentication features will stop working when Firebase Dynamic Links shuts down soon: email link authentication for mobile apps, as well as Cordova OAuth support for web apps.

Search by email address, phone number or user UID [Add user](#)

| Identifier          | Providers | Created     | Signed in   | User UID               |
|---------------------|-----------|-------------|-------------|------------------------|
| muervlath@gmail.com |           | 12 Jun 2026 | 12 Jun 2026 | Hxv3RzDy0r5d0h3pRm...  |
| marocwely@gmail.com |           | 11 Jun 2026 | 15 Jun 2026 | p0M6uUQv9d75m2wB4K5... |

Rows per page: 50 1 - 2 of 2

Type here to search

19°C

11:36 AM 6/12/2026

|     |                                                  |   |
|-----|--------------------------------------------------|---|
| AB- | <b>Agha Khan</b><br>Request • 2026-06-08         | ✓ |
| B+  | <b>Kenyatta hospital</b><br>Request • 2026-06-05 | ✓ |
| O+  | <b>Thika level 5</b><br>Request • 2026-06-05     | ✓ |

Hello, Maroakelly2



Search Blood Group



**Anna Itotiah W.**  
Nairobi



**Sharon Kendi.**  
Nakuru



**Faith Mueni.**  
Mombasa



**Roy Gichinga.**  
Kisumu



**Michelle Njogu.**  
Eldoret



**Kelley Maroa**  
Migori



**Nicole Chepkemboi**  
Nandi



Home



Search



Request



Profile

## Personal reflection

Learning about data storage and database integration helped me understand how mobile apps manage and persist user data efficiently. I now appreciate the importance of choosing the right database and ensuring secure, reliable data handling in application development.

WEEK 8: Handling user inputs and Events

20:39



## Welcome Back

Login to continue saving lives



Email Address

example@mail.com



Password



Maroakelly2  
@gmail.com

urgent

o



20:40



## Account Settings

### Appearance



Dark Mode



### Change Password

Old Password

New Password

Cancel

Save

Security



About



Version

1.0.0



20:39



## Welcome Back

Login to continue saving lives



Email Address

example@mail.com

Enter a valid email



Password



Password must be 6+ characters

Login

New donor? [Register Now](#)

20:40



Hello, Maroakelly1



 Nairobi, KE (Lat: -1.19)

## Quick Services



Find Donor



Emergency



Blood Bank



History

## Smart Match Suggestion



### Best Match Found!

Maroa Kelly (O+) is 500m away and available.



## Urgent Requests

[View All](#)

O+

Oasis  
Just now  
urgent!!



Home



Search



Request



Profile



20:40



Hello, Maroakelly1



Search Donor, Group or Location

mue



Faith Mueni

Machakos • AB-



muenifaith

Unknown • Unknown



mue

mueni

muen



20:45



← lib/main.dart



```
476         const SizedBox(height: 40),
477         Text(
478           "Welcome Back",
479           style: TextStyle(
480             fontSize: 32,
481             fontWeight: FontWeight.bold,
482             color: Colors.red.shade900,
483           ),
484         ),
485         const SizedBox(height: 10),
486         Text(
487           "Login to continue saving lives",
488           style: TextStyle(color: Colors
489             .grey.shade600, fontSize: 16),
490         ),
491         const SizedBox(height: 50),
492         TextFormField(
493           controller: emailController,
494           decoration: const InputDecoration(
495             labelText: "Email Address",
496             hintText: "example@mail.com",
497             prefixIcon: Icon(Icons
498               .email_outlined),
499             ),
500           keyboardType: TextInputType
501             .emailAddress,
502           validator: (v) => v!.contains('@') ?
503             null : "Enter a valid email",
504         ),
505         const SizedBox(height: 20),
506         TextFormField(
507           controller: passwordController,
508           obscureText: !isPasswordVisible,
509           decoration: InputDecoration(
510             labelText: "Password",
511             prefixIcon: const Icon(Icons
512               .lock_outline),
513             suffixIcon: IconButton(
514               icon: Icon(isPasswordVisible ?
```



Home



Inbox



Explore



Copilot

## Personal reflection

This topic helped me understand how mobile apps handle user input through events and gestures and how to do validation. I learned to structure functionality using classes, making applications more organized, interactive, and easier to maintain while improving user experience.

## WEEK 9: DEVICE FEATURES INTEGRATION



Welcome Back

Email Address

Face

Fingerprint

blood\_donation\_app



Authentication required

Verify identity

Access Life Flow Securely

Use PIN



Access

A



## ← Account Settings

### Security



**Change Password**

Update your login security



---

### Preferences

**Dark Mode**



**Biometric Security**

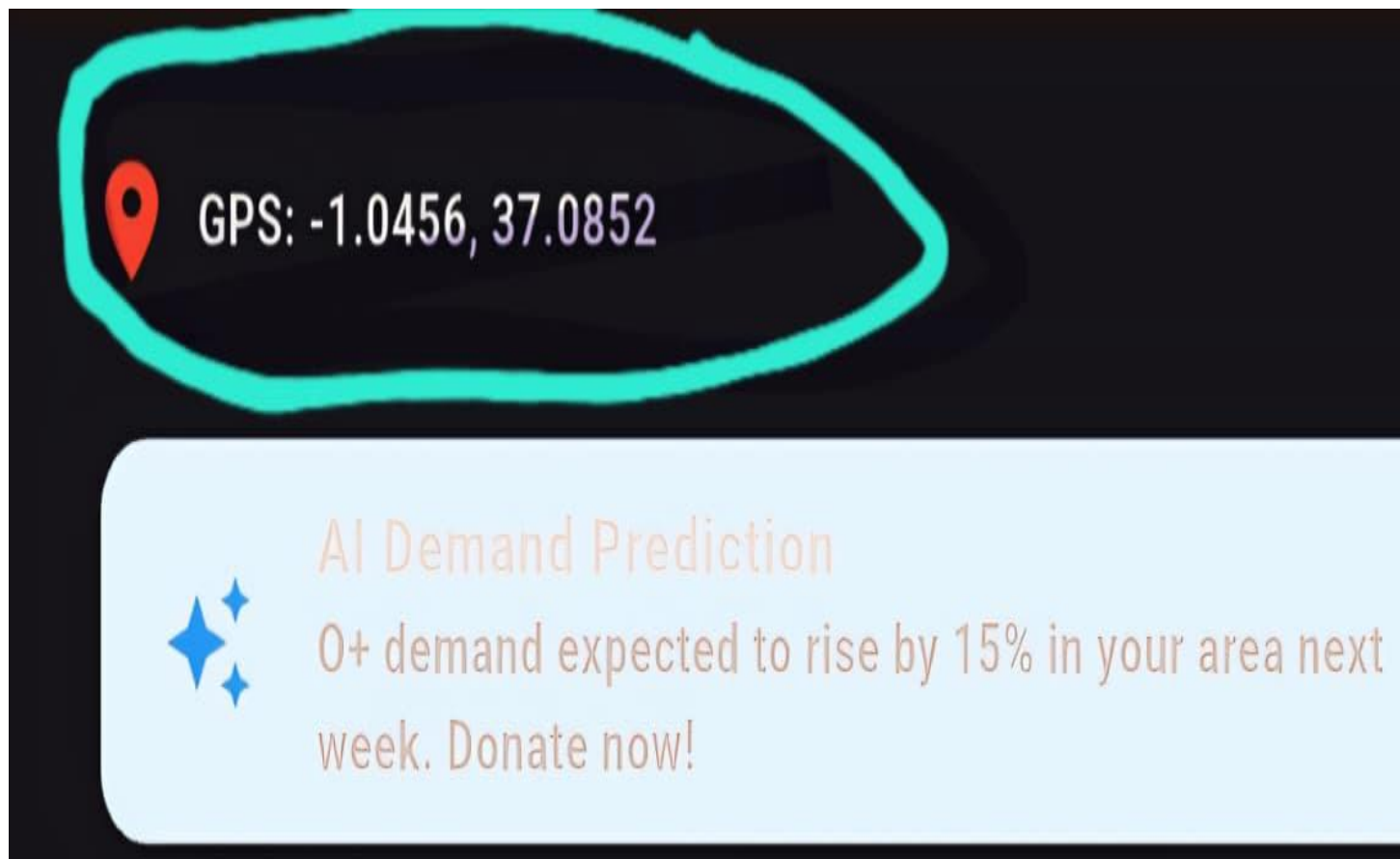
Use Fingerprint or Face ID



---

**App Version**

2.1.0



## PERSONAL REFLECTION

Through Week 9, I learned how to integrate device features like GPS and biometrics into my blood donation app. This improved my understanding of creating

interactive, user-friendly solutions while considering permissions, privacy, and real-world application efficiency.

## WEEK 10: COMPREHENSIVE APPLICATION DEVELOPMENT INTEGRATION AND TESTING

## Life Flow

### Create Blood Request

Blood Group Needed 



Hospital / Collection Point

Additional Message

Broadcast Request

Broadcast Successful!







20:21



LIFE FLOW

.

# Life Flow



Maroakelly1

Available to Donate



Accepting Requests

Toggle availability for blood requests



Donation Eligibility

Eligible to Donate

3

Donated

0

Points

9

Lives Saved

## Badges



First Gift



Saver



Verified



Donor ID Card





20:03



## Life Flow

Maroakelly1

Available to Donate



Accepting Requests

Toggle availability for blood requests



Donor ID



Badge



First Gift



Donor ID Card



Settings







## Account Settings

### Security



Change Password

Update your login security



### Preferences

Dark Mode



Biometric Security

Use Fingerprint or Face ID



App Version

2.1.0

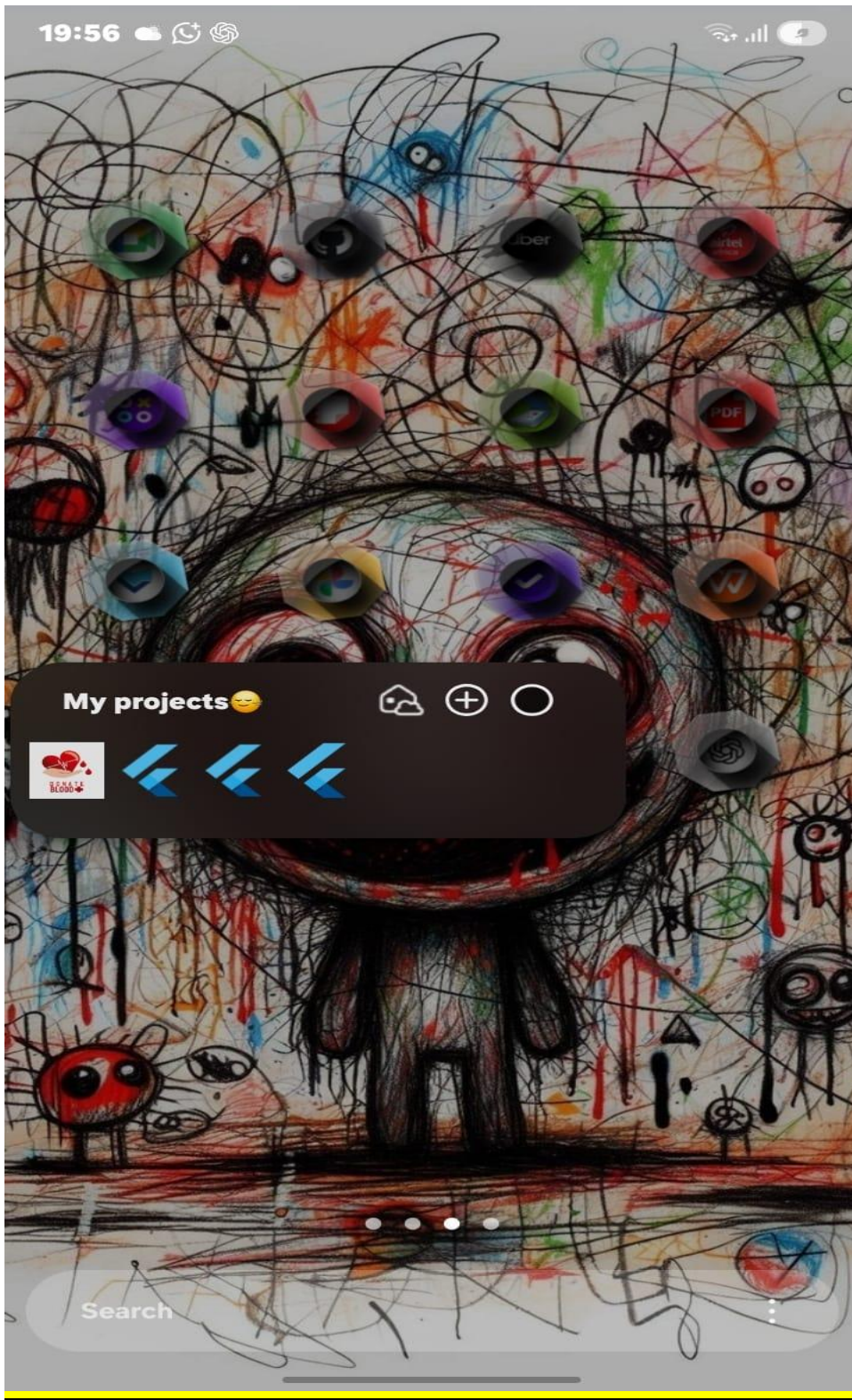
## PERSONAL REFLECTION

In Week 10, I learned how to integrate all the different components of a mobile application into one complete system. I improved my understanding of testing, debugging, and enhancing user experience. This helped me appreciate the importance of building a functional, reliable, and user-friendly application.

## **WEEK 11:APPLICATION DEPLOYMENT AND PUBLISHING**



19:56



My projects 😊



Search



19:56



LIFE FLOW

.

## BANNER



# BLOOD CONNECT

DONATE BLOOD. SAVE LIVES.

## YOUR DONATION CAN SAVE A LIFE

Blood Connect bridges the gap between donors and those in need. Together, we can build a healthier and stronger community.



Find Donors



Request Blood



Nearby Donors



Instant Notifications



Secure & Reliable





BE A HERO. DONATE BLOOD.

TOGETHER, WE SAVE MORE LIVES.

POWERED BY  **Firebase**

 Real-time Database

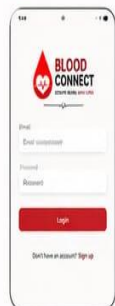
 Authentication

 Secure

## BLOOD CONNECT - USER GUIDE

A simple guide to help you navigate and make the most of the Blood Connect app.

### 1 REGISTER / LOGIN



Create a new account or login using your credentials.

### 2 DASHBOARD



Access all features from the dashboard. Choose what you need.

### 3 FIND DONORS



Search donors by blood group and location. Contact them directly.

### 4 REQUEST BLOOD



Fill in the details to request blood in emergencies.

### 5 MY REQUESTS



Track the status of your blood requests in real time.

### 6 NOTIFICATIONS



Stay updated with real-time alerts and important notifications.

### 7 MY PROFILE



Manage your profile, view donations, and update settings.

### TIPS

- ✓ Keep your profile information updated.
- ✓ Enable location for better donor results.
- ✓ Respond quickly to requests and messages.
- ✓ Donate blood regularly and encourage others.



ONE DONATION CAN SAVE THREE LIVES.

 Your data is safe with us. Blood Connect uses Firebase for secure authentication and real-time data synchronization.

 **Firebase**

 Authentication

 Realtime Database

 Cloud Functions

## ONE PAGE USER GUIDE

## BLOOD CONNECT APP – USER GUIDE

## **1. Introduction**

The Blood Connect App is designed to help users find blood donors quickly and request blood during emergencies. The application provides a simple and user-friendly interface for both donors and recipients.

## **2. Getting Started**

- Download and install the app.
- Open the app and register using your email and password.
- Log in to access the dashboard.

## **3. Key Features**

- Search Donors: Users can search for blood donors based on blood group and location.
- Request Blood: Users can submit emergency blood requests.
- Donor Registration: Users can register as blood donors.

- Profile Management: Users can update their personal details.

#### **4. How to Use the App**

- After logging in, navigate to the dashboard.
- Select “Search Donors” to find available donors.
- Use filters such as blood group to refine results.
- Click “Request Blood” to send an emergency request.
- Fill in the required details and submit.

#### **5. Notifications**

Users receive alerts when:

- A blood request is successfully submitted
- A donor responds to a request

#### **6. Safety and Privacy**

- User data is securely stored.
- Personal information is only used for donation purposes.

#### **7. Conclusion**

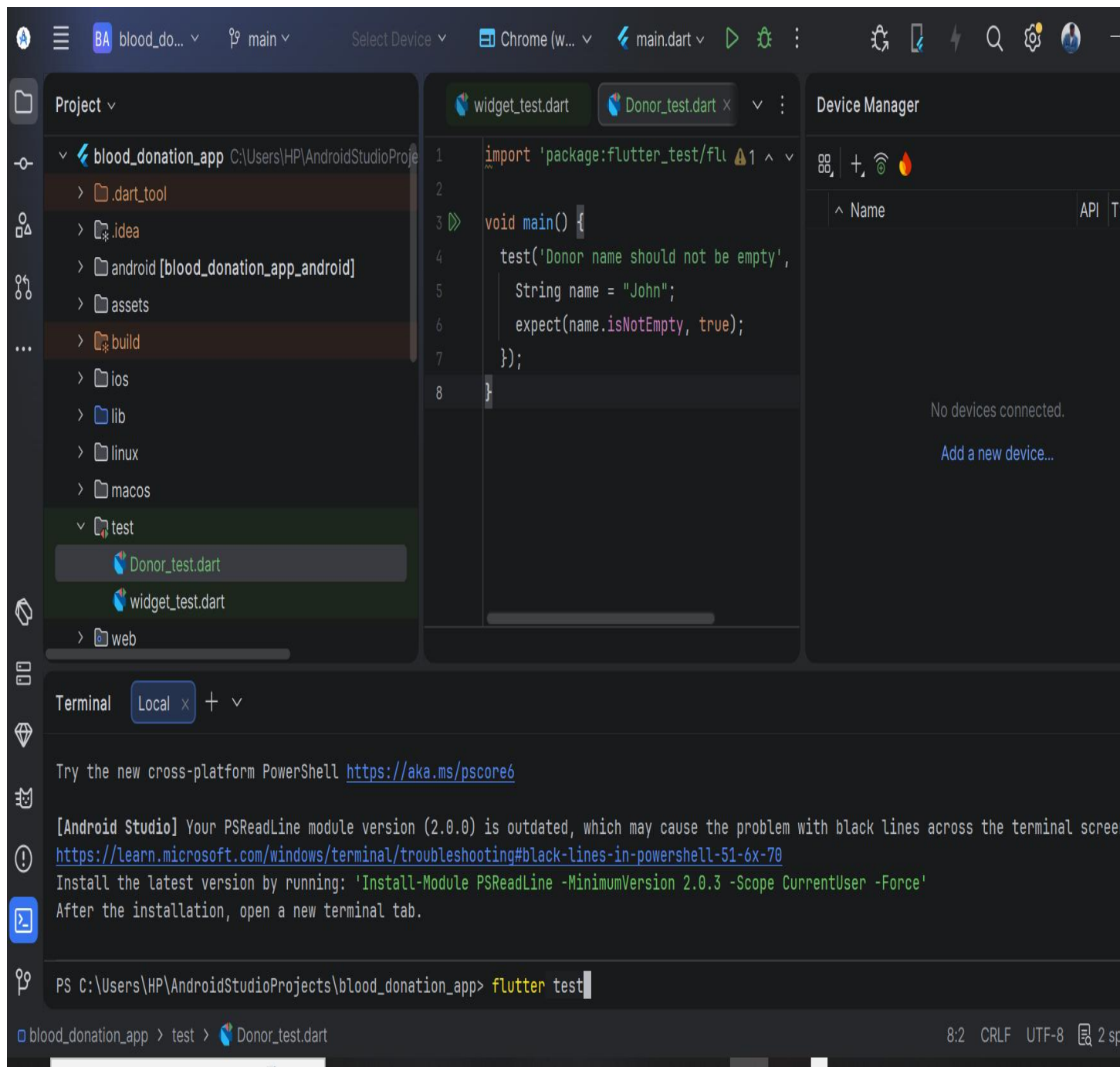
The Blood Connect App provides a fast, reliable, and efficient way to connect blood donors with recipients, helping save lives through technology.

## **PERSONAL REFLECTIONS**

Week 11 enhanced my understanding of application deployment. I learned to generate signed builds, prepare store assets, and perform final testing. It showed me that presentation, documentation, and quality assurance are essential before releasing an app to real users.

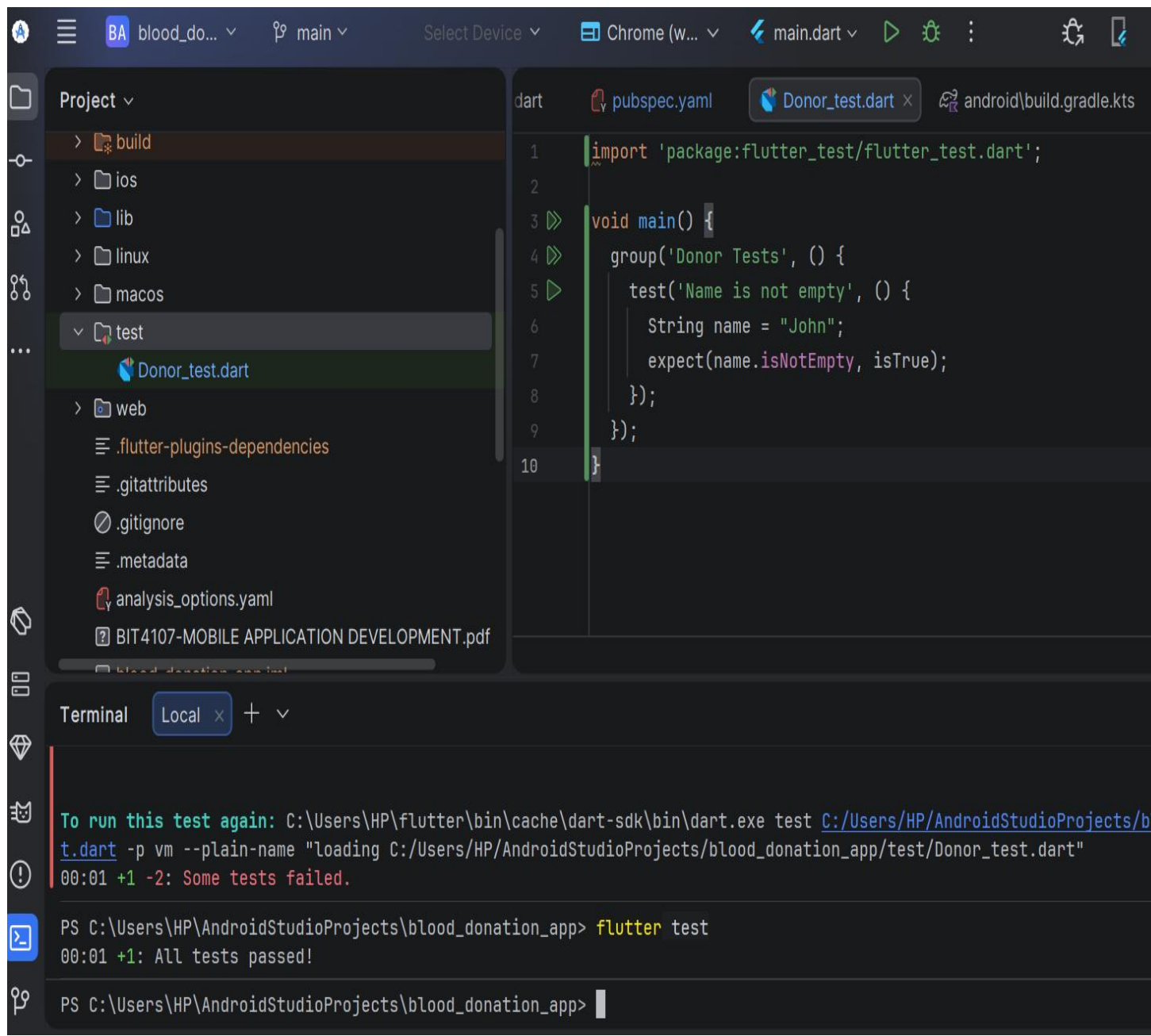
## **WEEK 12: TESTING AND DEBUGGING MOBILE APPLICATIONS**



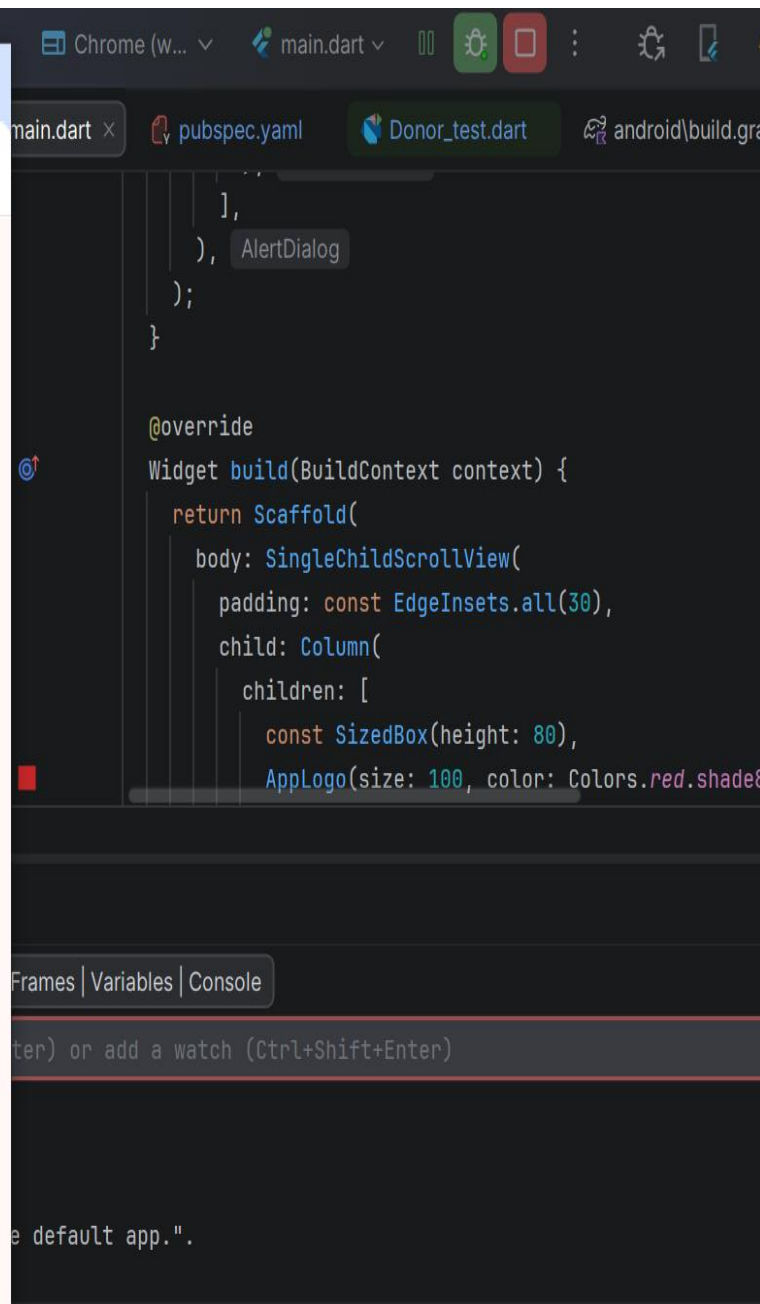
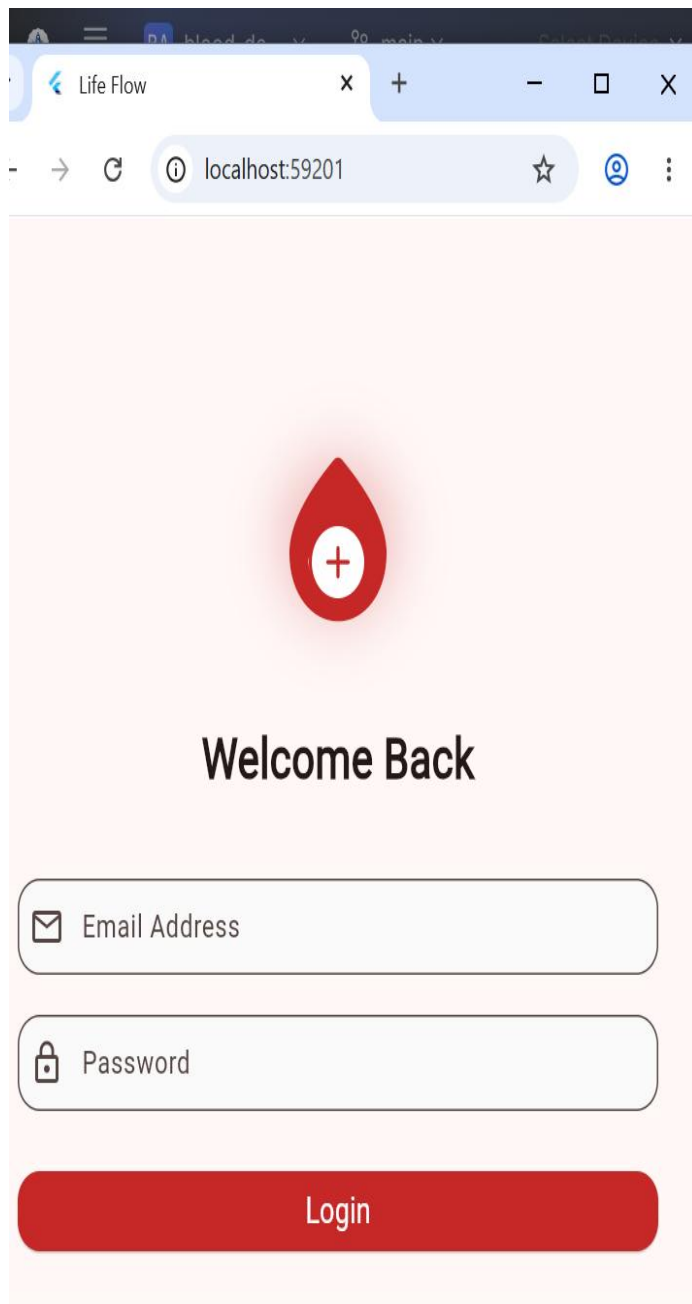


**Unit testing passed**





## Debug on chrome android studio



## Validation

20:49



# Welcome Back



Email Address

yyggh



Password

..|

Login



The email address is badly formatted.



## **Personal reflection**

I improved my understanding of Flutter DevTools and debugging. Faced issues connecting tools but learned how to navigate Inspector, Debugger, and Chrome DevTools. Gained confidence troubleshooting errors independently.

## **WEEK 13:PERFOMANCE OPTIMIZATION**

BA blood\_do...main

Select Device

Chrome (...)

main.dart

Project

.gitignore

.metadata

analysis\_options.yaml

BIT4107-MOBILE APPLICATION DEVELOPMENT.pdf

blood\_donation\_app.iml

devtools\_options.yaml

MOBILE DEVELOPMENT-BIT4107(DOCUMENTATION).pdf

pubspec.lock

pubspec.yaml

README.md

Week 1-6 version 2.pdf

week 1-7 version 3.pdf

Week 1-8 logbook.pdf

Week 1-10 logbook.pdf

External Libraries

Scratchpad and Console

main.dart

pubspec.yaml

Donor\_test.dart

android\build.gradle.kts

app\build.gradle.kts

514

515

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522

523

524

525

526

527

528

529

}

child:

), Elevat

l,

), AlertDialog

);

}

@override

Widget build(Buil

return Scaffold

body: SingleC

padding: co

child: Colu

children:

const S

Flutter DevTools

127.0.0.1:9100/home?compiler=...

Connected app

CPU / OS: Web windows

Connected app type: Flutter web (debug build)

VM Service Connection: ws://127.0.0.1:59693/PUIPc-7i9fY=/ws

Dart Version: 3.12.0

Flutter Version: 3.44.0 / stable

Framework / Engine: 559ffa3f75 / 4c525dac5e

Terminal

Local

c Clear the screen

q Quit (terminate the application on the device).

This app is linked to the debug service: ws://127.0.0.1:59693/PUIPc-7i9fY=/ws

Debug service listening on ws://127.0.0.1:59693/PUIPc-7i9fY=/ws

A Dart VM Service on Chrome is available at: http://127.0.0.1:59693/PUIPc-7i9fY=

The Flutter DevTools debugger and profiler on Chrome is available at:

http://127.0.0.1:59693/PUIPc-7i9fY=/devtools/?uri=ws://127.0.0.1:59693/PUIPc-7i9fY=

Starting application from main method in: org-dartlang-app:/web\_entrypoint.dart.

Firebase not initialized: Assertion failed:

Docs

Tutorial

Life Flow

127.0.0.1:9100/home?compiler=

127.0.0.1:9100/home?compiler=wasm&uri=ws://127.0.0.1:59693/PUiPc-7i9fY=/ws

Flutter DevToolsFlutter InspectorPerformanceDebuggerLoggingshared\_preferences

View VM flags

Connected app

CPU / OS: Web windows

Dart Version: 3.12.0

Connected app type: Flutter web (debug build)

Flutter Version: 3.44.0 / stable

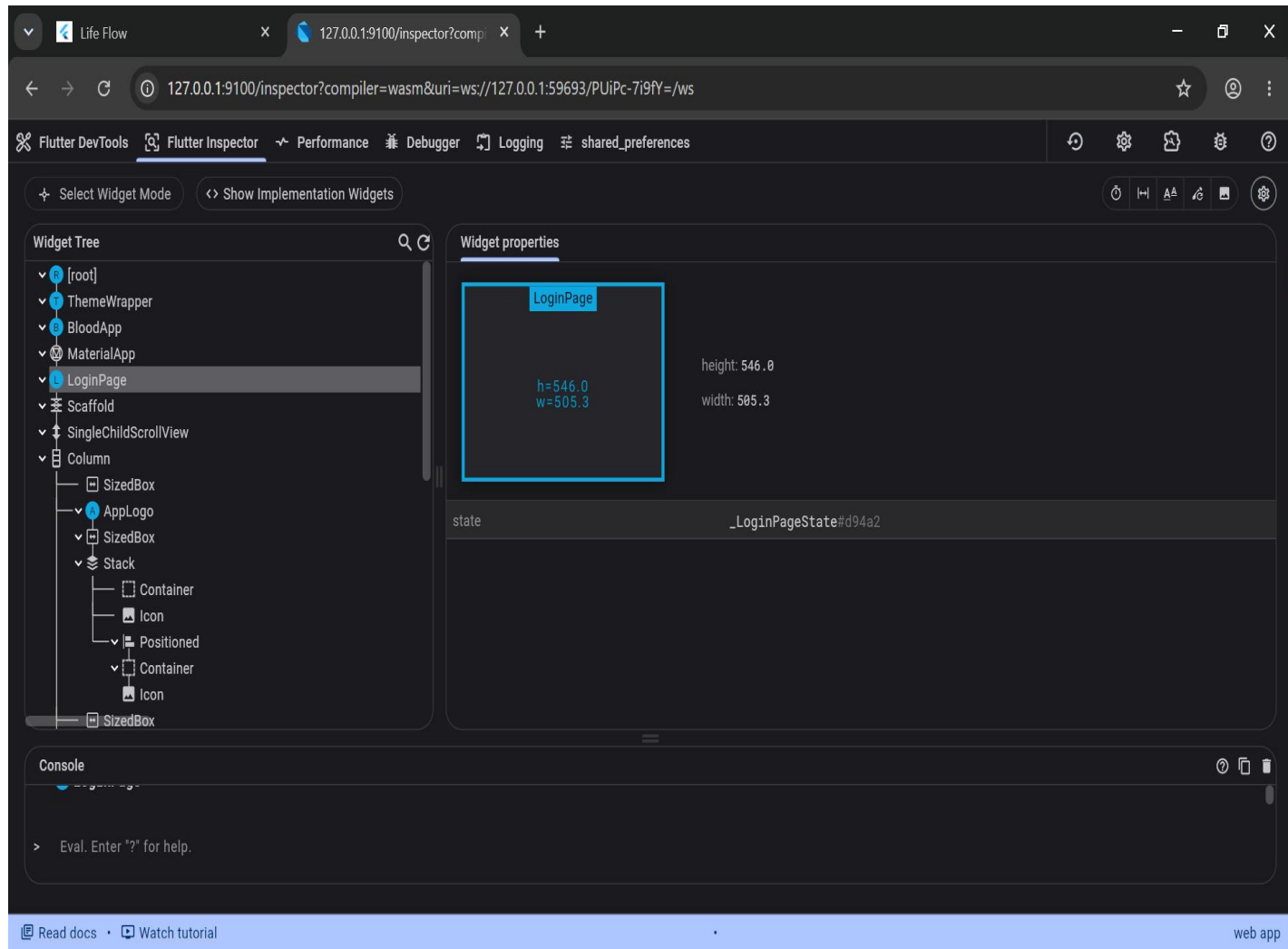
VM Service Connection: ws://127.0.0.1:59693/PUiPc-7i9fY=/ws

Framework / Engine: 559ffa3f75 / 4c525dac5e

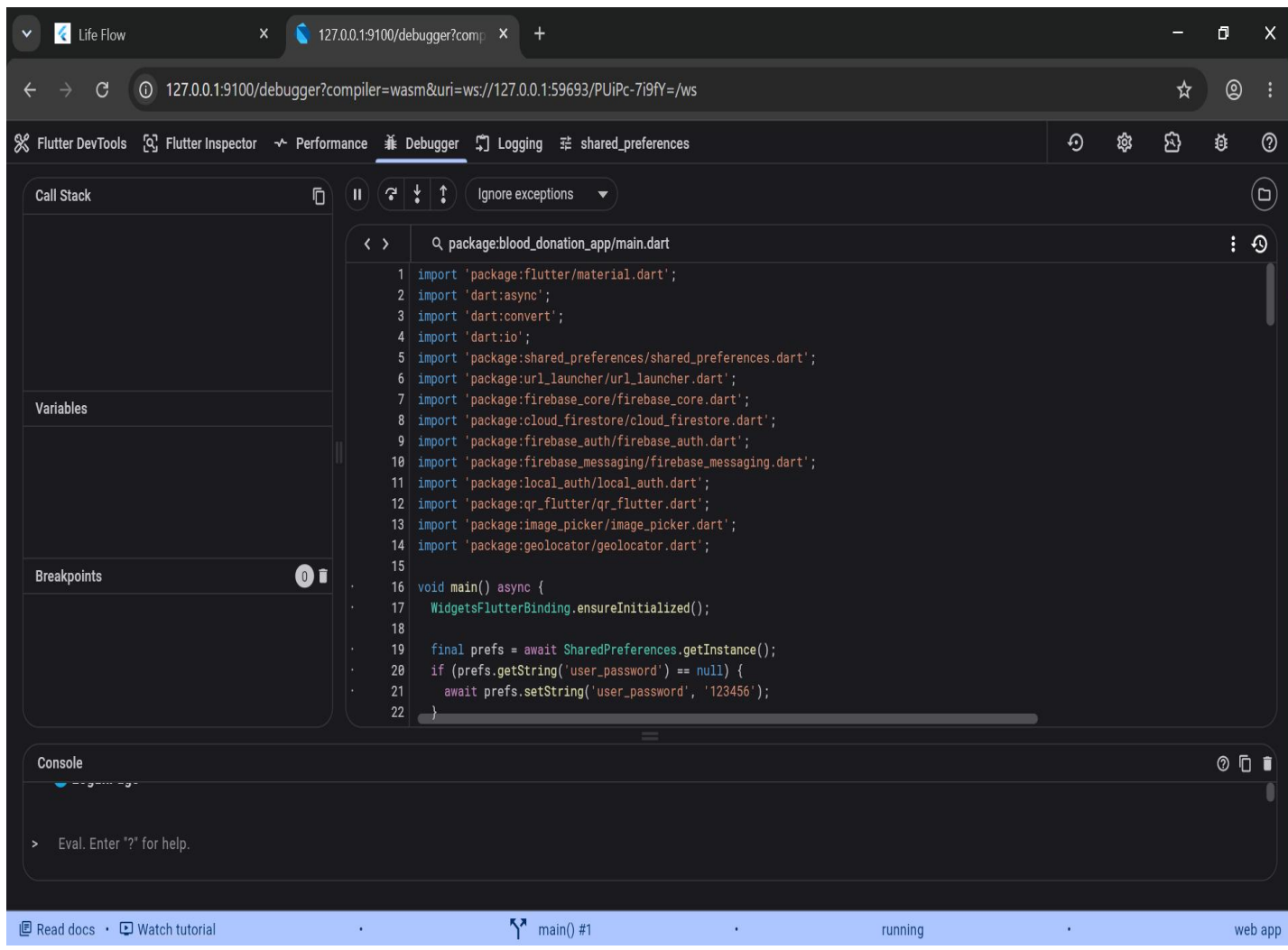
Read docs

Watch tutorial

# Flutter inspector



# debugger



performance



Life Flow

127.0.0.1:9100/performance?co

127.0.0.1:9100/performance?compiler=wasm&uri=ws://127.0.0.1:59693/PUiPc-7i9fY=/ws

☆ @

Flutter DevToolsFlutter InspectorPerformance>>

⌂⚙️🔍📄🔔

How to use Chrome DevTools for performance profiling

The Flutter framework emits timeline events as it works to build frames, draw scenes, and track other activity such as garbage collections. These events are exposed in the Chrome DevTools performance panel for debugging.

You can also emit your own timeline events using the `dart:developer` [Timeline](#) and [TimelineTask](#) APIs for further performance analysis.

Optional flags to enhance tracing

- `debugProfileBuildsEnabled`: Adds Timeline events for every Widget built.
- `debugProfileBuildsEnabledUserWidgets`: Adds Timeline events for every user-created Widget built.
- `debugProfileLayoutsEnabled`: Adds Timeline events for every RenderObject layout.
- `debugProfilePaintsEnabled`: Adds Timeline events for every RenderObject painted.

Instructions

- [Optional]* Set any desired tracing flags to true from your app's main method.
- Run your Flutter web app in [profile mode](#).
- Open up the [Chrome DevTools' Performance panel](#) for your application, and start recording to capture timeline events.

[Read docs](#) • [Watch tutorial](#)

ElementsConsoleSourcesPerformance>>⚠️8🔥1🔄⚙️⋮✕

🕒🔄🛑⬆️⬇️🏠Live metrics📷Screenshots⚙️

☐ Memory📊☐ Dim 3rd parties

Interaction to Next Paint (INP)

1,192 ms

Your local INP value of 1,192 ms is poor.

INP interaction [pointer](#)

[Learn more about local and field metrics](#)

InteractionsLayout shifts🔕

▶ keyboardinput.flt-text-editing.transparentTextEditing16 ms

▶ pointerflutter-view120 ms

▶ pointerflutter-view🕒 2,096 ms

Next steps

Field metrics

See how your local metrics compare to real user data in the [Chrome UX Report](#).

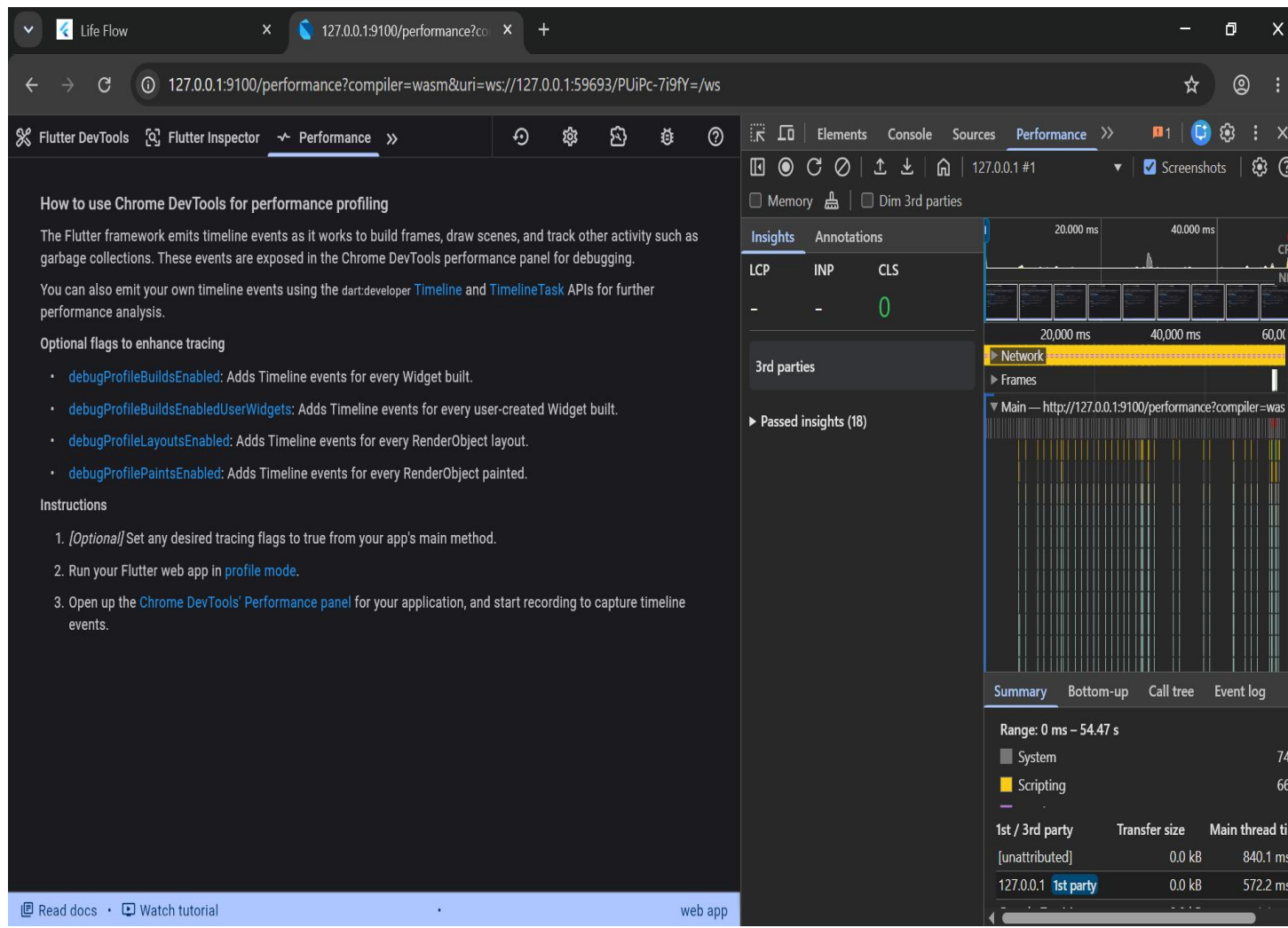
Set up

Environment settings

Use the [device emulation](#) and [performance throttling](#) to simulate real-world conditions.

web app

## Performance graph



## Personal reflection

Focused on performance profiling and optimizing my Flutter web app. Learned to use Chrome DevTools effectively. Began identifying slow UI components and improved overall app responsiveness.

## WEEK 14: FINAL CAPSTONE ASSIGNMENT

Enhanced problem-solving skills and workflow efficiency. Became more comfortable with debugging, profiling, and refining features. Reflected on progress and prepared for final improvements and project completion.